The empirical recurrence for OEIS A267456, proved from an exact model

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Abstract

OEIS A267456 carries an empirical recurrence of order 4 contributed by Colin Barker. It is true, and it is decidable rather than empirical. The entry's sequence is given exactly by a certified block template for the automaton's row read as a numeral, from which a provably satisfies a MONIC linear recurrence of order $S = 8$, derived below and not assumed. The residual of the conjectured recurrence satisfies the same one, so whether it annihilates a is settled by a finite exact computation. No transfer matrix is involved and the count is not a walk.

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1 The conjecture

OEIS A267456 is "Binary representation of the n-th iteration of the "Rule 133" elementary cellular automaton starting with a single ON (black) cell.". It has offset 0 and begins

$$1, 10, 100, 1000, 1010100, 10101000, 10101010100, 101010101000, \ \ldots$$

The entry states, as a conjecture contributed by its author and never marked settled:

> Conjectures from _Colin Barker_, Jan 15 2016 and Apr 19 2019: (Start) a(n) = 10001*a(n-2) - 10000*a(n-4) for n > 5.

As of the "Last modified" line on the live entry (Feb 16 2025 08:33:29, revision 28) this is still recorded as empirical, and nothing on the entry records it as proved.

2 The value is a certified block template

The entry reads the automaton's row at stage n as a string of digits and takes its value in base B. Nothing is being counted, so there is no digraph. The row does not grow only at its two ends — for these rules it gains digits in the MIDDLE of a periodic run — so the shape $w(n + p) = L_r + w(n) + R_r$ is unavailable. What holds instead is a block template: writing r for the residue of n modulo the period p and j for the number of periods past a pre-period n_0,

$$w(n_0 + r + jp) \ = \ B_0\, Q_1^j\, B_1 \cdots Q_m^j\, B_m,$$

with the fixed blocks B_i and the repeated blocks Q_i depending only on r. This was verified for every available n, and it is a theorem rather than a fit: the rule is a local map of radius 1, so p steps have radius p; the p-step map was checked by direct simulation at two consecutive j whose repeated runs are longer than $4p + 4|Q_i|$, and for a larger j the template only inserts further copies of each Q_i inside runs already longer than $2p$. An output cell near either end sees only input cells the two configurations share at the same distance from that end, and strictly inside a long run the input is $|Q_i|$-periodic so the output is too, with period and phase fixed by the two verified steps. Hence the template propagates to every j.

For this entry the automaton is rule 133, the period is $p = 2$ and the pre-period is $n_0 = 2$. The certificate is

$$\begin{aligned} r = 0: &\quad \mathsf{0} \cdot \mathsf{0101}^j \cdot \mathsf{0100} \\ r = 1: &\quad \mathsf{00} \cdot \mathsf{0101}^j \cdot \mathsf{01000} \end{aligned}$$

3 The bound

The row is the light cone, of width $2n+1$, so $\sum_i |Q_i| = 2p$. Write s_i for the number of digits to the right of the i-th repeated run, so $s_1 = 2p$. Concatenation is multiplication by a power of B and addition, and $v(Q_i^j) = v(Q_i)(B^{|Q_i|j} - 1)/(B^{|Q_i|} - 1)$, so along a residue class

$$a(j) \;=\; \sum_i \left(\alpha_i B^{s_i j} + \beta_i\right) + \gamma,$$

a $\mathbf{Z}$-linear combination of $B^{s_1 j}, \ldots, B^{s_m j}$ and 1 with coefficients that do not depend on j. In the shift $T = S^p$ that is annihilated by $\prod_i (T - B^{s_i})(T - 1)$. Taking the DISTINCT roots that occur across the p residue classes, pulling each back to $z^p - \lambda$, and allowing the pre-period to raise the numerator's degree gives the monic annihilator of order $S = 8$ used below.

4 The proof

Lemma 1. *Let $A(z) = z^S - \sum_{j<S} \alpha_j z^j$ be the monic annihilator derived above and $q(z) = z^4 - \sum_i c_i z^{4-i}$ the characteristic polynomial of the conjectured recurrence. The residual $r(n) = a(n) - \sum_i c_i a(n-i)$ is a linear combination of shifts of a, so A annihilates r as well. Since $A(0) \neq 0$ the recurrence it gives runs in both directions, and S consecutive zeros of r force r to vanish at every later index.*

Proof. A annihilates a, hence every shift of a, hence any linear combination of shifts; that is r. A monic recurrence with nonzero constant term determines each term from the S before it and each from the S after it, so a run of S zeros propagates. ☐

Theorem 2.

$$a(n) \;=\; 10001\, a(n-2) - 10000\, a(n-4)$$

for every $n > 5$.

Proof. The terms $a(0), \ldots$ were computed exactly in integer arithmetic from the model, extended by A where the model itself was not evaluated, and $r(n)$ formed for each index. Those integers vanish from $n = 5+1$ onwards, and the run exceeds $S+4$, which by Lemma 1 settles every larger index at once. The bound is exact: at $n = 5$ the recurrence fails on the entry's own published terms, so no larger range holds. ☐

5 Verification

Three checks, each independent of the algebra above.

First, the model reproduces all 16 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry: it is built by reading the entry's English, and a misreading gives different counts.

Second, the conjectured recurrence was evaluated directly on those published terms, with no model involved, and holds wherever the proved range and the data overlap.

Third, the derived annihilator A was itself tested on terms it had not been given: the model was evaluated beyond the S values that determine it and A reproduced them. A bound that is too small fails this test, which is why it is run.

References

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